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1. We may say that a graph $G = (V, E)$ is *topologically disconnected* if we can find two sets $U, W \subset V$ such that:

1. $U \cup W = V$,
2. $U \cap W = \emptyset$,
3. Both U and W are non-empty, and
4. There are no edges in G connecting a vertex in U to a vertex in W .

(The phrase “topologically connected” is *not* standard mathematical terminology.) Prove that a graph is connected if and only if it is not topologically disconnected.

Hint: Proof by contradiction.

2. Let $G = (V, E)$ be a graph. A set of vertices $I \subseteq V$ is called an *independent set* if there are no edges between any pair of vertices in I . Namely, $x, y \in I$ implies $(x, y) \notin E$. Let C be the size of the largest independent set in G . Prove the chromatic number $\chi(G)$ satisfies

$$\chi(G) \geq \frac{|V|}{C}.$$